House Prices and School Choice:

Evidence from Chicago's Magnet Schools Proximity Lottery

Leonardo Bonilla
Department of Economics
University of Illinois
lbonill2@illinois.edu

Esteban Lopez
Department of Urban and Regional Planning
University of Illinois
lopezoch@illinois.edu

Daniel McMillen
Department of Economics
University of Illinois
mcmillen@illinois.edu

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Abstract

Studies of open school policies suggest that prices rise in areas served by low-quality schools when the students gain access to high-quality schools outside of their local district. However, excess demand may lead some students to be denied admission to high-quality schools. We take advantage of changes in admission policies to Chicago's magnet schools to test whether a higher probability of admission to high-quality schools leads to higher house prices. Chicago adopted policies that increased the probability of admission for students living within 1.5 miles of a magnet school. A comparison of house prices on either side of the 1.5-mile contour suggests that the increase in admission probabilities due to the 1997 reform increased house prices for homes within the 1.5-mile radius by as much as 4.6%, and the premium is still higher for homes in areas near multiple magnet schools. The 2009 reform is estimated to have produced a premium of approximately 13% for homes within the 1.5-mile radius by allowing homes near magnet schools to avoid some of the dramatic drop in house prices that began at the end of 2007. Quantile estimates suggest that the reforms had the largest effect on moderate-priced homes.

1. Introduction

School choices are typically tied to residential location decisions in the United States. Particularly at the elementary school level, enrollments are apt to be restricted to students living within in relatively small neighborhoods near the school. One result of these restrictive enrollment policies is that households who value education will be willing to pay a premium to live in districts with high-quality schools. The empirical literature on the capitalization of school quality into house prices is sufficiently large to have generated two recent reviews (Machin, 2011 and Nguyen-Hoang and Yinger, 2011), both of which conclude that households are willing to pay a significant premium to live in neighborhoods with schools whose students have high test scores.

In contrast to these closed enrollment policies, many districts offer a form of open enrollment. Students may have the option to attend any school within their district or sometimes even in another school district. The theoretical literature on open enrollment policies (e.g., Epple and Romano, 2003; Ferreyra, 2007; and Nechyba, 2000, 2003) suggests that house prices will rise in areas with lower-quality schools, while house prices decline in areas whose schools receive large numbers of outside students. Reback (2005) found evidence supporting both predictions in a study of Minnesota school districts following the adoption of an inter-district open enrollment policy. Brunner, Cho, and Reback (2012) also find evidence supporting these predictions using data from 12 states that had adopted inter-district choice programs as of 1998. Analyses of intra-district open enrollment policies reach similar conclusions: prices rise in areas with low-quality schools and prices fall in areas with high-quality schools (Machin and Salvanes, 2010; Schwartz, Voicu, and Horn, 2014).

The question addressed in these papers is whether the presence of school choice is capitalized into house values. Choices are more valuable if they can clearly be granted. In some

districts, access to high-quality schools is limited by enrollment caps. A mechanism is required to allocate enrollment when there is excess demand for a school. Local students typically get priority, while out-of-district may be determined by some form of lottery.

In this study, we analyze the effects of school choice on house values within a single, large school district, the Chicago Public School (CPS) district. Chicago designated a set of magnet schools in response to a desegregation order in 1980. Although any Chicago student could potentially enroll, a citywide lottery system was used to grant admission. The lottery included minority quotas, but students living near a school were not given priority. A reform affecting only elementary schools was introduced in December 1997 assigning higher probabilities of admission to students who lived within 1.5 miles of a magnet elementary school. Higher probability of admission was also granted to students with a sibling already attending the school. Another round of reform was introduced in December 2009 after the desegregation order was rescinded. The 2009 reform increased the percentage of seats that could be assigned based on proximity to the school from 30% to 40%, and it removed the restriction on the number of seats that could be assigned to siblings. The effect of these reforms was to significantly increase the probability of admission for a student living within 1.5 miles of a magnet elementary school.

These two reforms serve as a natural experiment allowing us to determine whether the higher probability of potential admission is capitalized into house prices. Following an approach introduced by Black (1999) and followed by a host of subsequent authors, we compare house prices on either side of the 1.5-mile boundary to determine whether the reforms altered home values significantly.[1] We find that the 1997 increased property values within the 1.5-mile zones

[1] Example of studies using geographical discontinuities to estimate causal effects of schools on house values include Bayer, Ferreira, and McMillan (2007); Bogart and Cromwell (2000); Fack and Grenet (2010); Gibbons and Machin (2003); Gibbons, Machin, and Silva (2013); Ries and Somerville (2010); and Schwartz, Voicu, and Horn (2014).

by at nearly 5%. The 2009 form had an even larger effect of about 13%, although in this recessionary period the main effect was to greatly reduce the amount by which prices fell. Quantile estimates suggest that these effects are most pronounced for homes with prices in the middle of the distribution. While the entire distribution of prices shifts to the right after both 1997 and 2009 for homes with 1.5 miles of a magnet school, the shift is relatively small at the tails of the distribution.

Although our study's main contribution to the literature is to establish that geographically-based admission probabilities have a significant effect on house prices, our results also have implications for the literature on school quality. Magnet schools are high-quality schools, and although admission is not restricted to high-achieving students, the students who are attracted to magnet schools are those who place a high value on education. Thus, our study supports recent work by such authors as Brasington and Haurin (2006); Clapp, Nanda, and Ross (2008); Hoxby (2004); Kane, Staiger, and Samms (2003); Kane, Rieg, and Staiger (2006); and Rouse and Barrow (2009) showing that school quality and academic performance affects property values.

2. Chicago's Magnet Schools

The history of Chicago's magnet school system is discussed in Allensworth and Rosenkranz (2000). The CPS established Chicago's magnet schools in response to a 1980 desegregation consent decree signed with the federal government. The original goal of the decree response was to increase the percentage of white students in the CPS from its low level in 1980 (less than 20%) by establishing a set of high-quality schools that would attract white students. Since Chicago's neighborhoods are highly racially segregated, neighborhood-based school admissions produce racially segregated schools. The magnet schools had racial quotas ranging

from 15% - 35% white. "The hope was that by offering special schools, children from all over the city would be attracted to them. Thus, a multiracial student body could be achieved in some schools in a system that had far too many racially isolated schools due to the housing pattern segregation that existed (and still exists) in Chicago" (Allensworth and Rosenkranz, 2000, p. 7).

Elementary magnets classified as "regular magnets" were created in direct response to the consent decree and were subject to the racial quotas. Another set of schools was created that were not subject to the quotas, including "scholastic academies", "regional gifted centers", "classical schools", and "academic centers". Of these, all but the scholastic academies were limited to high-achieving students. The categories of magnet high schools are similar: "traditional magnets" were subject to racial quotas, while "regional college preparatory schools" and "international baccalaureate programs" had no quotas but were limited to high-achieving students. As admission policies for categories of magnets other than regular elementary magnets have not changed significantly over time and continue to be citywide, only regular magnets are included in our empirical analysis, and the remainder of the section focuses on this category of magnet school.

Prior to 1997, a general, citywide lottery was conducted to allocate admissions for all magnet schools, including the regular magnets. A result of the excess demand for magnet school enrollment was that students living near a school might be denied enrollment. In response, the CPS now conducts a "proximity lottery" that reserves a portion of the enrollment slots for students living within 1.5 miles of a regular magnet school. The proximity lottery was announced in December 1997 and was implemented for the 1998 – 1999 school year. In the first school year, 15% of the enrollment slots were reserved for the neighborhood. The percentage has been 30% since the 2000 – 2001 school year. The neighborhood is defined precisely using straight-line distance from the student's address to the school.

The proximity lottery still leaves excess demand for many regular magnet schools. To assure that siblings can attend the same school, beginning in 1997 45% of the enrollment slots were set aside for siblings of students who already attend the school. Finally, 5% of the enrollment slots were reserved for allocation at the principal's discretion. The combination of the proximity lottery and the provision for siblings provides a strong incentive for families with children to choose a home in an area within 1.5 miles of a regular magnet. Moreover, some regions fall within the requisite 1.5 miles of as many as 5 magnets. Since families can enter the proximity lottery for all schools with 1.5 miles of their home, these regions are especially valuable.

In September 2009, a federal court decision vacated the desegregation consent decree.[2] In December 2009, the Chicago Board of Education approved a new admissions policy for magnet schools. The most significant changes were the elimination of race-based admissions criteria and the removal of the restriction that no more than 45% of the seats were reserved for siblings. After all siblings are enrolled, 40% of the remaining seats are now reserved for students living within 1.5 miles of a regular magnet school, regardless of race. Proximity lotteries are conducted if the number of neighborhood applicants exceeds 40% of the available seats. However, race and ethnicity continues to matter. According to the *Chicago Public Schools Policy Manual*, "In an effort to ensure ongoing diversity in these programs, if more than 50% of the entire student body, according to the 20th day file, is comprised of students within the proximity and if more than 50% of the student body is any one racial or ethnic group, no proximity lottery will be held for that school."[3]

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[2] According to the CPS web site, whites currently account for 9.4% of enrollment. Hispanics now form the largest group, with 45.6% of total enrollment. African-Americans comprise 39.3% of total enrollment, which stood at 234,679 for elementary schools and 112,029 for secondary schools in Fall 2014. These figures are drawn from http://cps.edu/About_CPS/At-a-glance/Pages/Stats_and_facts.aspx.

[3] http://policy.cps.k12.il.us/download.aspx?ID=82.

Students who are not admitted as a sibling or via the proximity lottery can still gain admission to a magnet school through the Citywide SES Lottery. "SES" is an acronym for "socio-economic status." A score for socio-economic status is assigned to each census tract based on six criteria: median family income, adult educational attainment, the percentage of single-parent households, the percentage of home ownership, the percentage of the population that speaks a language other than English, and a school performance variable. The school performance variable is based on ISAT scores for schools with attendance areas in the census tract. The *Chicago Public Schools Policy Manual* includes the following summary of the Citywide SES Lottery":

> "Lotteries will be conducted within each of the four SES tiers and applicants will be ranked in lottery order within each tier. If there are insufficient applicants within a tier to fill the allocated number of seats in that particular SES tier, the unfilled seats will be divided evenly and redistributed across the remaining tier(s) as the process continues. A sufficient number of offers will be made in lottery order for each SES tier to fill the seats allocated to this lottery process. The remaining applicants will be placed on an applicant wait list by SES tier." http://policy.cps.k12.il.us/download.aspx?ID=82, p. 4.

The new admission policy was implemented for the 2010-2011 school year. The effect of removing the racial quotas, increasing the proximity lottery share to 40%, and expanding the number of seats allocated to siblings is to provide a strong incentive for families with children to live in areas that are within 1.5 miles of one or more magnet schools. Even if a student ultimately attends a private or parochial school, owning a home near a magnet school may serve as a form of insurance for parents that their child will not be forced to attend a low-quality school. Thus, house prices can be expected to rise for homes that are within 1.5 miles of a magnet school.

3. Data

Our primary data source is DataQuick, which provided data on house sales for Chicago for 1997-2012. We then expanded the sample to include data from 1993-1996 using data from the Illinois Department of Revenue. DataQuick also provided information on lot size, building area, house age, the number of bathrooms, and indicators that the home has central air conditioning, brick construction, or a fireplace. Comparable data for 1993-1996 are drawn from the Cook County Assessor's Office. As DataQuick relies on the same data sources when constructing their sales files, the two sets of data are directly comparable. Most of our analysis focuses on two sub-periods encompassing the reform dates – 1995-2000 and 2007-2012. Focusing on these sub-periods helps to isolate the effects of the reforms and avoids complications arising from the booming housing markets of 2003-2006.

We restrict the sample to sales of Class 2 homes that are within 3 miles of a regular magnet school that had opened before 1998. The Cook County Assessor's Office defines Class 2 properties as residential buildings with 6 units or fewer. Condos are also excluded from the analysis because we do not have data on structural characteristics for them. Table 1 provides the list of magnet schools included in the analysis, and Figure 1 shows their locations within the city.

Descriptive statistics for the two sub-periods of data are presented in Table 2. Averages for most variables are similar for homes in areas affected by the reforms and for homes located more than 1.5 miles from a magnet school.

4. Empirical Approach

Following Black (1999) and much of the subsequent literature, we use a difference in differences approach to estimate the effect of the admission reforms on house prices. Letting lnP_{hst} represent the log sale price of home *h* in school district *s* at time *t*, the basic estimating equation is

$$lnP_{hst} = \gamma_1 Treat_{ht} + \gamma_2 Treat_{ht} x Reform_t + X_{hs}\beta + \mu_s + \rho_t + u_{hst}$$

when evaluating data from one of the two sub-periods, 1995 – 2000 and 2007 – 2012. We also pool data across the two sub-periods and estimate the following equation:

$$lnP_{hst} = \gamma_1 Treat_{ht} + \gamma_2 Treat_{ht}\ x\ Reform97_t + \gamma_3 Treat_{ht}\ x\ Reform09_t + X_{hs}\beta + \mu_s + \rho_t + u_{hst}$$

We include fixed effects for the quarter of sale and the census tract, and standard errors are clustered at the tract level.[4] The estimating equations also include controls for structural characteristics, including log building area, log lot size, building age, the number of bathrooms, and dummy variables for central air conditioning, brick construction, and a fireplace.[5]

Two reforms took place during our sample period, one at the end of 1997 and the other at the end of 2009. As the geographic area covered by the reforms does not differ over time, the *Treat* variable is the same for both reform times: *Treat* = 1 if a home is within 1.5 miles of a magnet school. To evaluate the December 1997 reform, we estimate models using data for 1995 – 2000, while we use data from 2007 – 2012 to analyze the December 2009 reform. For the 1995 –

[4] The results are similar when elementary school districts are used as the basis for geographic fixed effects rather than census tracts. Census tracts are smaller than school districts: for our sample of sales of homes that are within 1.5 miles of a magnet school, there are 817 census tracts and 339 elementary school districts.

[5] Implicitly, the estimating equations also include controls for the *Reform97* and *Reform09* variables. However, these variables are not separately identified from the controls for quarter of sale.

2000 data, $Reform_t$ = 1 for $t \geq$ 1998:1, while $Reform_t$ = 1 for $t \geq$ 2009:1 for the 2007 – 2012 data. When the two data sets are pooled, the same rules are used to define $Reform97_t$ and $Reform09_t$.

We vary the size of the band around the critical 1.5 mile distance mark by estimating models with the sample restricted to sales of homes located in a band ranging from 1.5 – δ to 1.5 + δ miles of a magnet school. Thus, *Treat* = 1 if the home lies within 1.5 – δ of a magnet school and *Treat* = 0 if the home is in the 1.5 + δ range. We present results for three values of δ: 1.5, 1, and 0.5. Figure 2 is an illustration of this process. Smaller values of δ lead to fewer observations for both the treatment and control groups as the sample is restricted to a narrower band around the critical 1.5 distance. Observations are likely to be more similar across the treatment and control groups within narrower bands around the discontinuity point: in the limit, the rule would match only neighboring houses on either side of the 1.5 mark. The housing characteristic variables and census tract fixed effects control for heterogeneity introduced by having the larger samples produced by wider distance bands. Figure 3 shows the treatment and control observations when δ = 1.5, while Figure 4 shows the much smaller number of observations when δ = 0.5.

A sizable portion of the sample is located in areas of the city that are within 1.5 miles of more than one magnet school. Figure 5 illustrates a representative case where some observations are close to two schools. Table 3 shows the number of observations that are within a 1.5 miles of 0-4 schools for each of the three distance bands. Even when δ = 0.5, more than 15% of the treatment observations are within 1.5 miles of 2 or more magnet schools. To measure the effects of treatment intensity – the number of nearby magnet schools – on house prices, we add separate $Treat$ and $Treat\ x\ \ Reform$ variables for observations that are close to 2 or 3-4 schools.

5. Results

Figure 6 shows the path of median house prices over time for the control and treatment observations defined by the three distance bands. The vertical lines indicate the start of the first quarter following the December 1997 and December 2009 reform dates, i.e., 1998:1 and 2010:1. In each case, median prices start at nearly identical levels in 1993, but a wedge forms before the 1997 reform, after which median prices are significantly higher for properties closer to the magnet schools. The treatment premium does not vary greatly over the subsequent decade. The 2010 reform appears to have averted some of the collapse in house prices that began in late 2007. Prices fell much more for control properties than for homes close to magnet schools. Although prices rose again for control properties after 2010, the discount for control properties remains larger at the end of 2012 than it had been in earlier years.

Table 3 presents our primary regression results. All sales within 3 miles of a magnet school are included in this set of regressions, i.e., $\delta = 1.5$. The regressions include controls for the quarter of sale and census tract fixed effects. Standard errors are clustered by census tract.

The results for the structural characteristics are standard. Sales prices are estimated to be higher for bigger, newer homes on larger lots. Prices are also higher for homes with more bathrooms, brick construction, and a fireplace. The only anomaly is the negative sign for central air conditioning in the 1995-2000, but the estimated coefficients turns to the expected positive value in the later time period.

The key results are listed last in Table 4. The results for 1995-2000 indicate that prices rose by approximately 4.6% in areas that had admission probabilities increased by the 1997 reform. The results for 2007-2012 indicate that the effect was of the 2009 was larger at 13.4%. When the

two time periods are pooled, the estimate for the effect of the 1997 reform falls to 2.9% while the effect of the 2009 reforms rises to 19.8%. All estimates are highly statistically significant.

Table 5 presents the results for treatment intensity. For both time periods, the interactions between the number of nearby magnets and $Treat\ x\ \ Reform$ imply a higher treatment effect for homes that are within 1.5 miles of a larger number of magnet schools. The results for the 1995 – 2000 period imply that house prices rose by 2.6% after the 1997 reform for homes that are within 1.5 miles of one magnet school, by 10.4% for homes that are near two schools, and by 13.0% for homes that are near three or four schools. Comparable figures for the 2009 reform are 11.5%, 8.5%, and 36.3% for homes that are within 1.5 miles of 1, 2, or 3-4 schools.

Table 6 shows the effect of varying δ. The results are generally quite stable as the band narrows around the 1.5 mile distance. In the specification that does not distinguish between the number of nearby schools, the coefficient for $Treat\ x\ \ Reform97$ falls from 0.046 to 0.029 as δ declines from 1.5 to 0.5, while the coefficient for $Treat\ x\ \ Reform09$ declines from 0.134 to 0.089. The results for the specification in which the coefficients vary by the number of nearby magnets schools suggest that much of this decline is due to the differences in the effects by treatment intensity. The coefficients for $Treat\ x\ \ Reform\ x\ Near\ 1\ Magnet\ School$ decline as δ declines, becoming statistically insignificant for the 1998 reform. However, the coefficients tend to rise as δ declines for 2 and 3-4 magnet schools.

6. Placebo Tests

In this section, we report the results for a set of placebo tests. First, we use an incorrect definition of the treatment area: any home within 0.5 miles of a magnet school is defined as having received the treatment of a higher probability of admission. With $\delta = 0.5$, this treatment definition means that both "treatment" and "control" observations have actually been beneficiaries of the reforms. Thus, we should not expect to find statistically significant estimates for the $Treat\ x\ \ Reform$ variables. The results are shown in Table 7. As expected, none of the estimated coefficients for $Treat\ x\ \ Reform97$ or $Treat\ x\ \ Reform09$ are statistically significant.

As another check on the accuracy of our models, we estimate a set of regressions with an incorrect definition of the treatment date. We define the treatment date as December 1994 rather than December 1997 and restrict the sample to sales from 1993-1996. In this case, none of the observations has received a higher probability of admission. The results are shown Table 8. As expected, the incorrect treatment is indicated to have no effect on house prices.

Our final placebo test employs an incorrect definition of the category of school having a proximity lottery. Instead of regular magnets, we make the incorrect assumption that the seven "scholastic academies" that were open prior to 1997 received higher admission probabilities.[6] Although these seven schools are similar to the regular magnet schools in many ways, they did not adopt proximity lotteries for admission. We then repeat our analysis by comparing sales prices on either side of the 1.5-mile contour around the seven scholastic academies. The results are shown

[6] The scholastic academies without proximity lotteries that were open in 1997 are Decatur Classical Elementary School, the Edison Elementary Regional Gifted Center, Keller Elementary Gifted Magnet School, Lenart Elementary Regional Gifted Center, McDade Elementary Classical School, Poe Elementary Classical School, and Skinner North Elementary School.

in Table 9. Again, none of the estimated coefficients for $Treat\ x\ Reform97$ or $Treat\ x\ Reform09$ are statistically significant, and the coefficients for 1997 have the wrong sign.

7. Quantile Regression Results

Although the magnet schools have proved popular, they did not achieve the objective of attracting white students to the Chicago public school system. High-income families from all ethnic backgrounds are apt to send their children to private schools. Magnet school admission policies may be a minor concern to such families, as well as to households without children. If so, the admission reforms may have more effect on relatively low-priced housing.

We estimate a series of quantile regressions to determine whether the effects of the reforms vary across the distribution of house prices. With 817 census tracts and 24 quarters in each six-year time interval, some simplifications are necessary to make estimation feasible. We follow a two-stage estimation procedure proposed by Canay (2011) to control for census tract fixed effects. The first stage is our basic linear regression model, which includes controls for both census tracts and the quarter of sale. The first-stage regression provides estimates of the census tracts fixed effects, $D\alpha$, where D is the $n\ x$ 817 matrix of census tract indicator variables. Under the assumption that α does not vary across quantiles, the second stage is a standard quantile regression of $lnP - D\hat{\alpha}$ on *Treat*, *Treat x Reform*, *X*, and controls for the time of sale. As a further simplification, we use a cubic function to control for time in the second stage quantile regressions: $\sum_{j=1}^{3}\{\delta_j(q-q_1)^j + \lambda_j(q-q_1)^j xReform\}$, where q represents the quarter of sale. This specification allows for different time trends before and after the reform dates.

Table 10 presents the quantile regression results for the *Treat* and *Treat x Reform* variables for three quantiles, τ = 0.10, 0.50, *and* 0.90. For 1990-2000, the estimated coefficients for *Treat x Reform97* are larger for the low and high quantiles than for the median.[7] In contrast, the estimated coefficient for *Treat x Reform09* is higher for the median than for τ = 0.10 and 0.90. Models 3 and 4 allow the effects to vary by treatment intensity. As expected, the estimated treatment effects are higher for homes in areas that are near more magnet schools. For the 1995-2000 data, the estimated effect of the 1998 reform is higher for the median than for τ = 0.10 and 0.90 when a home is within 1.5 miles of two magnet schools, while the estimates are higher for larger quantiles when the home is near 3-4 magnet schools. For 2007-2012, the estimated treatment effects are always estimated to be higher for the median than at the 10% and 90% quantiles.

To better illustrate the effects of the reforms on the full distribution of house prices, we estimate a series of quantile regression for quantile ranging from τ = 0.02 to τ = 0.98 in increments of 0.01. At each quantile, we predict the sale price at the time just after the reform using the following equation:

$$ln\hat{P} = \hat{\gamma}_1(\tau)Treat + \hat{\gamma}_2(\tau)TreatxReform + X\hat{\beta}(\tau)$$

$$+\sum_{j=1}^{3}\{\hat{\delta}_j(\tau)(q-q_1)^j + \hat{\lambda}_j(\tau)(q-q_1)^j xReform\}$$

with *X* set at the actual values in the data set and q = 13, i.e., the first quarter after the reform. To analyze the effects of the reform on the treated observations (*Treat* = 1), we compare the predictions for observations within 1.5 miles of a magnet school for $Reform = 0$ and $Reform =$

[7] Standard errors are reported conditional on the first-stage estimates and are not clustered by census tract.

1. The result is a set of $n_1 x$ 97 predicted values for $Reform = 0$ and $Reform = 1$, where n_1 is the number of observations with *Treat* = 1. We then estimate kernel density functions using the full set of entries for each matrix. A comparison of the estimated kernel densities shows the shift in the full distribution of house prices for the treated observations due to the reform.

The results are shown in Figure 8. For 1995-2000, the distribution of house prices shifts to the right by a relatively small amount. The shift is somewhat larger on the left side of the distribution than at high prices. The shift is much larger for 2007-2012. The result of the 2009 reform is to shift the distribution well to the right, particularly for prices near the middle to middle-left portion of the distribution.

The procedure can also be used to simulate the effect of the reforms for alternative treatment intensities. The $Treat$ and $TreatxReform$ are interacted with the variables *NumSchools* indicating whether the treated observation is within 1.5 miles of 1, 2, or 3-4 schools. We again estimate quantile regression for τ = 0.02, 0.03, … , 0.98. The next step is to calculate the full set of $n_1 x$ 97 predictions for $Reform = 0$ and $Reform = 1$ for *NumSchools* = 1, 2, and 3-4. The results are shown in Figure 9. For 1995-2000, the effect of the 1997 reform is now seen to be quite pronounced for homes near 2 or 3-4 magnet schools. The 2009 reform is estimated to produce a still larger rightward shift in the price distribution. For both reforms, the shifts are concentrated in the middle to middle-left portion of the price distribution.

8. Conclusion

The literature on school choice has largely neglected the need to ration spaces in high-quality schools. Having the right to apply for admission to a school does not guarantee a student a seat. The option may well be valuable, as suggested by the studies by Reback (2005), Cho, and Reback (2012), Machin and Salvanes (2010), and Schwartz, Voicu, and Horn (2014), all of which suggest that house prices rise in areas that gain open enrollment in high-quality schools. However, the premium should be higher if students have a lower probability of being denied a seat in the desirable schools.

We take advantage of changes in admission policies to magnet schools to test whether a higher probability of admission to high-quality schools leads to higher house prices. Chicago's magnet schools were created in response to a 1980 desegregation consent decree. Although regular magnet schools did not restrict admission to high-achieving students, there soon was an excess demand for seats as they gained a reputation as high-quality schools. At the end of 1997, Chicago introduced a proximity lottery that increased the probability of admission to students living within 1.5 miles of a magnet school. When the consent decree expired in 2009, Chicago again increased admission probabilities for students living within this radius by removing racial quotas, eliminating a restriction on the proportion of a school's enrollment devoted to siblings, and increasing the proportion of the seats allocated to students living within the 1.5-mile radius.

Using data on house sales for 1993 – 2012, we find strong evidence that these admission reforms increased prices for homes within 1.5 miles of a magnet school as compared to homes in neighboring areas that did not benefits from the reforms. Prices are estimated to have increased by as much as 7.5% as a result of the 1997 reform. The premium is still higher – in the 10% - 20% range – for the subset of homes in areas for which admission probabilities rose dramatically as a

result of being within the 1.5-mile radius of more than two schools. The 2009 reform is also estimated to have a large effect on house prices, with homes within the 1.5-mile radius earning a premium of approximately 20% over more distant housing. The 2009 appears to have helped homes within 1.5 miles of a magnet school avoid some of the dramatic drop in house prices that occurred during the late 2000s. Quantile estimates suggest that the price increases were most pronounced for homes in the middle to middle-left portion of the price distribution.

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Table 1

List of Magnet Schools Operating in 1997

Albert R Sabin Elementary Magnet School
Andrew Jackson Elementary Language Academy
Burnside Elementary Scholastic Academy
Edward Beasley Elementary Magnet Academic Center
Frank W Gunsaulus Elementary Scholastic Academy
Franklin Elementary Fine Arts Center
Galileo Math & Science Scholastic Academy Elementary School
Hawthorne Elementary Scholastic Academy
Inter-American Elementary Magnet School
Jensen Elementary Scholastic Academy
John H Vanderpoel Elementary Magnet School
LaSalle Elementary Language Academy
Leif Ericson Elementary Scholastic Academy
Maria Saucedo Elementary Scholastic Academy
Mark Sheridan Elementary Math & Science Academy
Ole A Thorp Elementary Scholastic Academy
Robert A Black Magnet Elementary School
Stone Elementary Scholastic Academy
Turner-Drew Elementary Language Academy
Walt Disney Magnet Elementary School
Walter L Newberry Math & Science Academy Elementary School
William Bishop Owen Scholastic Academy Elementary School

Table 2

Descriptive Statistics

	Within 1.5 Miles of a Magnet School			1.5 – 3 Miles from a Magnet School		
	1995 - 2000	2007 - 2012	1995 - 2012	1995 - 2000	2007 - 2012	1995 - 2012
Log of Sale Price	11.847 (0.710)	12.058 (1.108)	12.130 (0.865)	11.639 (0.615)	11.644 (1.085)	11.865 (0.807)
Log of Building Area	7.434 (0.507)	7.453 (0.505)	7.447 (0.507)	7.327 (0.472)	7.355 (0.467)	7.344 (0.473)
Log of Lot Size	8.158 (0.351)	8.114 (0.400)	8.131 (0.377)	8.248 (0.271)	8.219 (0.297)	8.228 (0.285)
Age/10	7.287 (3.187)	8.205 (3.673)	7.697 (3.506)	7.322 (2.462)	8.597 (2.800)	7.992 (2.714)
Bathrooms	2.070 (1.104)	2.207 (1.171)	2.158 (1.140)	1.810 (0.937)	1.919 (0.972)	1.883 (0.971)
Central Air Conditioning	0.238 (0.426)	0.274 (0.446)	0.262 (0.440)	0.159 (0.365)	0.177 (0.382)	0.170 (0.375)
Brick	0.611 (0.488)	0.611 (0.488)	0.605 (0.489)	0.557 (0.497)	0.536 (0.499)	0.536 (0.499)
Fireplace	0.113 (0.317)	0.101 (0.301)	0.102 (0.302)	0.072 (0.258)	0.067 (0.250)	0.066 (0.249)
Within 1.5 Miles of 1 Magnet School	0.755 (0.430)	0.693 (0.461)	0.724 (0.447)	0	0	0
Within 1.5 Miles of 2 Magnet Schools	0.150 (0.357)	0.203 (0.402)	0.179 (0.383)	0	0	0
Within 1.5 Miles of 3-4 Magnet School	0.095 (0.293)	0.104 (0.305)	0.097 (0.296)	0	0	0
Number of Observations	48,683	35,217	153,095	51,799	38,367	168,624

Note. Standard deviations are in parentheses.

Table 3

Number Observations by Number of Nearby Magnet Schools

Number of Magnet Schools within 1.5 Miles	1995-2000	2007-2012	1995-2012
$\delta = 1.5$			
0	130,257	38,367	168,624
1	86,436	24,411	110,847
2	20,279	7,151	27,430
3	10,101	3,298	13,399
4	1,062	357	1,419
Total	248,135	73,584	321,719
$\delta = 1.0$			
0	97,969	29,015	126,984
1	74,105	21,265	95,370
2	15,701	5,500	21,201
3	6,005	2,041	8,046
4	1,062	357	1,419
Total	194,842	58,178	253,020
$\delta = 0.5$			
0	52,513	15,693	68,206
1	44,687	12,920	57,607
2	5,642	1,946	7,588
3	2,030	713	2,743
4	335	101	436
Total	105,207	31,373	136,580

Table 4

Estimated Differences in Differences Effects on House Prices

Variable	1995-2000	2007-2012	1995-2012
Log of Building Area	0.337 (0.006)	0.298 (0.010)	0.334 (0.005)
Log of Lot Size	0.254 (0.011)	0.346 (0.017)	0.248 (0.010)
Age/10	-0.024 (0.001)	-0.044 (0.002)	-0.024 (0.001)
Bathrooms	0.010 (0.003)	0.018 (0.004)	0.014 (0.002)
Central Air Conditioning	-0.012 (0.005)	0.060 (0.007)	0.009 (0.004)
Brick	0.054 (0.004)	0.099 (0.008)	0.071 (0.004)
Fireplace	0.061 (0.007)	0.064 (0.011)	0.064 (0.006)
Within 1.5 Miles of a Magnet School	-0.001 (0.016)	-0.025 (0.021)	-0.024 (0.019)
Within 1.5 Miles of a Magnet School, Post-1997	0.046 (0.009)		0.029 (0.014)
Within 1.5 Miles of a Magnet School, Post-2009		0.134 (0.023)	0.198 (0.031)
R^2	0.725	0.747	0.728
Number of Observations	100,482	73,584	321,719

Notes. Clustered standard errors are in parentheses. The dependent variable is the log of sale price. The regressions include controls for the quarter of sale and census tract fixed effects.

Table 5

Estimated Effect on House Prices by Treatment Intensity

Variable	1995-2000	2007-2012	1995-2012
Log of Building Area	0.338 (0.006)	0.299 (0.010)	0.335 (0.005)
Log of Lot Size	0.253 (0.011)	0.346 (0.017)	0.247 (0.010)
Age/10	-0.024 (0.001)	-0.044 (0.002)	-0.024 (0.001)
Bathrooms	0.010 (0.002)	0.018 (0.004)	0.014 (0.002)
Central Air Conditioning	-0.012 (0.005)	0.061 (0.007)	0.009 (0.004)
Brick	0.054 (0.004)	0.099 (0.008)	0.071 (0.004)
Fireplace	0.062 (0.007)	0.064 (0.010)	0.064 (0.006)
Within 1.5 Miles of a 1 Magnet School	0.010 (0.016)	-0.019 (0.021)	-0.009 (0.020)
Within 1.5 Miles of a 2 Magnet Schools	-0.014 (0.027)	0.070 (0.032)	-0.021 (0.035)
Within 1.5 Miles of a 3-4 Magnet Schools	-0.080 (0.037)	-0.043 (0.048)	-0.109 (0.028)
Within 1.5 Miles of a 1 Magnet School, Post-1997	0.026 (0.010)		0.014 (0.015)
Within 1.5 Miles of a 2 Magnet Schools, Post-1997	0.104 (0.020)		0.071 (0.028)
Within 1.5 Miles of a 3-4 Magnet Schools, Post-1997	0.130 (0.013)		0.098 (0.017)
Within 1.5 Miles of a 1 Magnet School, Post-2009		0.115 (0.023)	0.174 (0.032)
Within 1.5 Miles of a 2 Magnet Schools, Post-2009		0.085 (0.038)	0.130 (0.051)
Within 1.5 Miles of a 3-4 Magnet Schools, Post-2009		0.363 (0.048)	0.488 (0.057)
R2	0.726	0.748	0.729
Number of Observations	100,482	73,584	321,719

Notes. Clustered standard errors are in parentheses. The dependent variable is the log of sale price. The regressions include controls for structural characteristics, the quarter of sale and census tract fixed effects.

Table 6

Variation in Distance Bands

Variable	δ = 1.5	δ = 1.0	δ = 0.5
1995-2000 Data			
Within 1.5 Mile of Magnet School, Post-1997	0.046 (0.009)	0.041 (0.010)	0.029 (0.012)
2007-2012 Data			
Within 1.5 Mile of Magnet School, Post-2009	0.134 (0.023)	0.130 (0.024)	0.089 (0.031)
1995-2012 Data			
Within 1.5 Miles of 1 Magnet School, Post-1997	0.014 (0.015)	0.012 (0.015)	0.003 (0.016)
Within 1.5 Miles of 2 Magnet Schools, Post-1997	0.071 (0.028)	0.063 (0.031)	0.083 (0.042)
Within 1.5 Miles of 3-4 Magnet Schools, Post-1997	0.098 (0.017)	0.125 (0.024)	0.154 (0.032)
Within 1.5 Miles of 1 Magnet School, Post-2009	0.174 (0.032)	0.188 (0.034)	0.130 (0.042)
Within 1.5 Miles of 2 Magnet Schools, Post-2009	0.130 (0.051)	0.175 (0.057)	0.218 (0.077)
Within 1.5 Miles of 3-4 Magnet Schools, Post-2009	0.488 (0.057)	0.437 (0.076)	0.260 (0.122)

Notes. Clustered standard errors are in parentheses. The dependent variable is the log of sale price. The regressions include controls for structural characteristics, the quarter of sale, and census tract fixed effects.

Table 7

Treatment Defined as within 0.5 Miles, $\delta = 0.5$

Variable	1995-2000	2007-2012	1995-2012
Within 1.5 of a Magnet School	-0.003 (0.017)	-0.022 (0.023)	0.009 (0.025)
Within 1.5 of a Magnet School, Post-1997	0.007 (0.019)		-0.017 (0.024
Within 1.5 of a Magnet School, Post-2009		0.044 (0.035)	0.019 (0.044)

Notes. Clustered standard errors are in parentheses. The dependent variable is the log of sale price. The regressions include controls for structural characteristics, the quarter of sale, and census tract fixed effects.

Table 8
Reform Date Defined as 1994

	$\delta = 1.5$	$\delta = 1.0$	$\delta = 0.5$
Within 1.5 of a Magnet School	0.011 (0.015)	0.011 (0.015)	0.017 (0.014)
Within 1.5 of a Magnet School, Post-1994	0.011 (0.006)	0.010 (0.006)	-0.002 (0.008)

Notes. Clustered standard errors are in parentheses. The dependent variable is the log of sale price. The regressions include controls for structural characteristics, the quarter of sale, and census tract fixed effects. The sample includes data from 1993-1996.

Table 9

Seven Scholastic Academies with Selective Admission Standards

Variable	$\delta = 1.5$	$\delta = 1.0$	$\delta = 0.5$
Within 1.5 Miles of a Magnet School	0.039 (0.026)	0.022 (0.027)	0.023 (0.027)
Within 1.5 Miles of a Magnet School, Post-1997	-0.050 (0.015)	-0.027 (0.017)	-0.031 (0.018)
Within 1.5 Miles of a Magnet School, Post-2009	0.055 (0.041)	0.028 (0.044)	0.057 (0.048)

Notes. Clustered standard errors are in parentheses. The dependent variable is the log of sale price. The regressions include controls for structural characteristics, the quarter of sale, and census tract fixed effects. The sample includes data from 1995-2012.

Table 10
Quantile Regression Results

Variable	$\tau = 0.10$	$\tau = 0.50$	$\tau = 0.90$
Model 1: 1995-2000			
Within 1.5 Miles of a Magnet School	0.008 (0.012)	0.012 (0.004)	0.020 (0.006)
Within 1.5 Miles of a Magnet School, Post-1997	0.028 (0.015)	0.019 (0.005)	0.024 (0.008)
Model 2: 2007-2012			
Within 1.5 Miles of a Magnet School	0.013 (0.012)	-0.002 (0.009)	-0.035 (0.014)
Within 1.5 Miles of a Magnet School, Post-2009	0.051 (0.018)	0.100 (0.014)	0.070 (0.020)
Model 3: 1995-2000			
Within 1.5 Miles of 1 Magnet School	0.019 (0.012)	0.018 (0.004)	0.015 (0.007)
Within 1.5 Miles of 2 Magnet Schools	-0.042 (0.031)	0.004 (0.010)	0.102 (0.017)
Within 1.5 Miles of 3 Magnet Schools	-0.131 (0.052)	-0.059 (0.017)	0.018 (0.028)
Within 1.5 Miles of 1 Magnet School, Post-1997	0.019 (0.016)	0.009 (0.005)	0.009 (0.009)
Within 1.5 Miles of 2 Magnet Schools, Post-1997	0.062 (0.040)	0.119 (0.013)	0.045 (0.022)
Within 1.5 Miles of 3 Magnet Schools, Post-1997	0.104 (0.064)	0.116 (0.021)	0.129 (0.035)
Model 4: 2007-2012			
Within 1.5 Miles of 1 Magnet School	0.021 (0.013)	-0.002 (0.010)	-0.048 (0.014)
Within 1.5 Miles of 2 Magnet Schools	0.070 (0.030)	0.093 (0.023)	0.094 (0.033)
Within 1.5 Miles of 3 Magnet Schools	0.039 (0.046)	0.072 (0.035)	0.108 (0.050)
Within 1.5 Miles of 1 Magnet School, Post-2009	0.046 (0.019)	0.089 (0.015)	0.069 (0.021)
Within 1.5 Miles of 2 Magnet Schools, Post-2009	0.016 (0.044)	0.153 (0.033)	0.152 (0.048)
Within 1.5 Miles of 3 Magnet Schools, Post-2009	0.196 (0.067)	0.232 (0.051)	-0.038 (0.073)

Notes. Clustered standard errors are in parentheses. The dependent variable is the log of sale price. The regressions include controls for structural characteristics, time trends, and census tract fixed effects. The sample includes data from 1995-2012.

Figure 1

Magnet School Locations

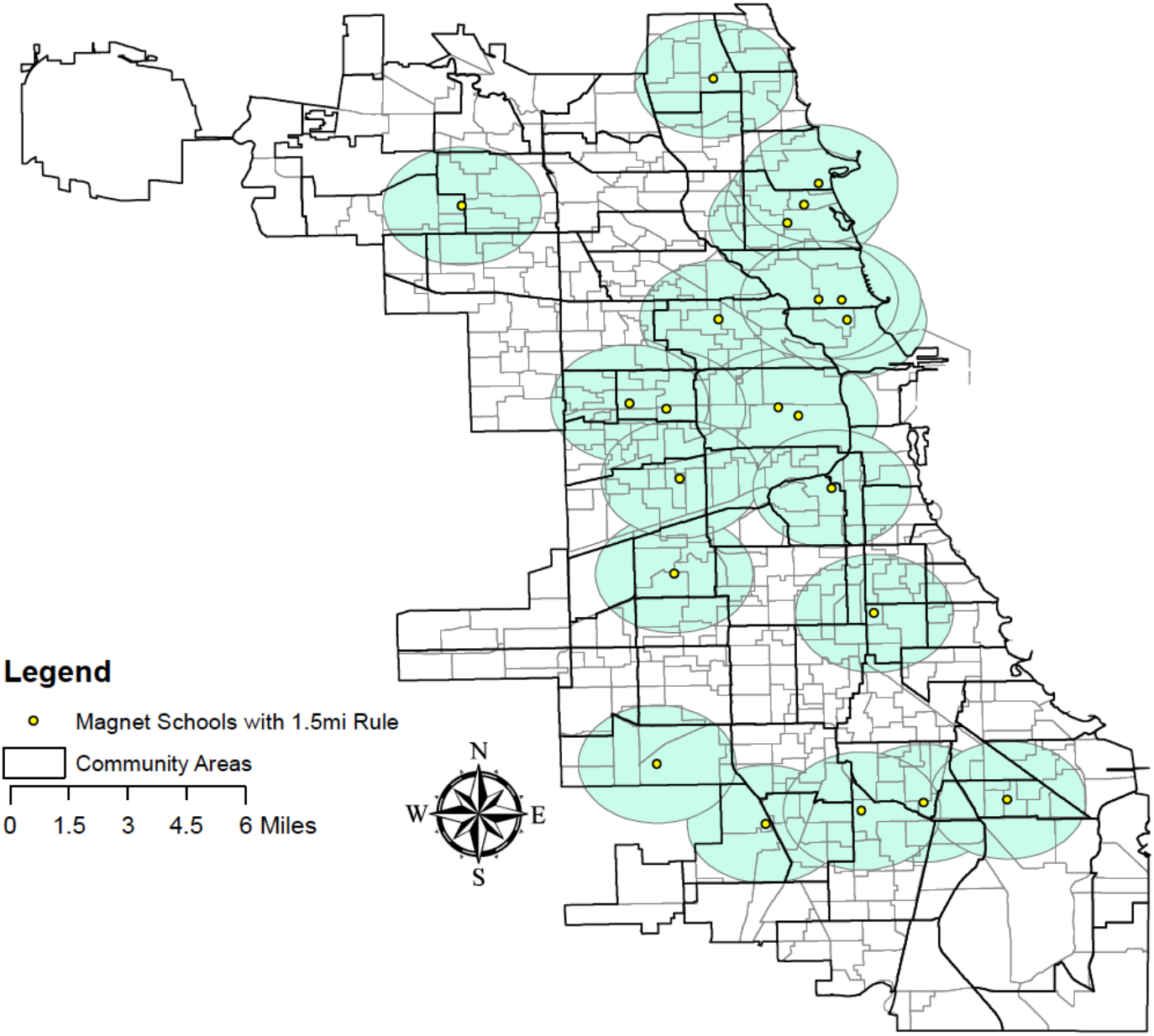


Figure 2

Treatment and Control Definitions

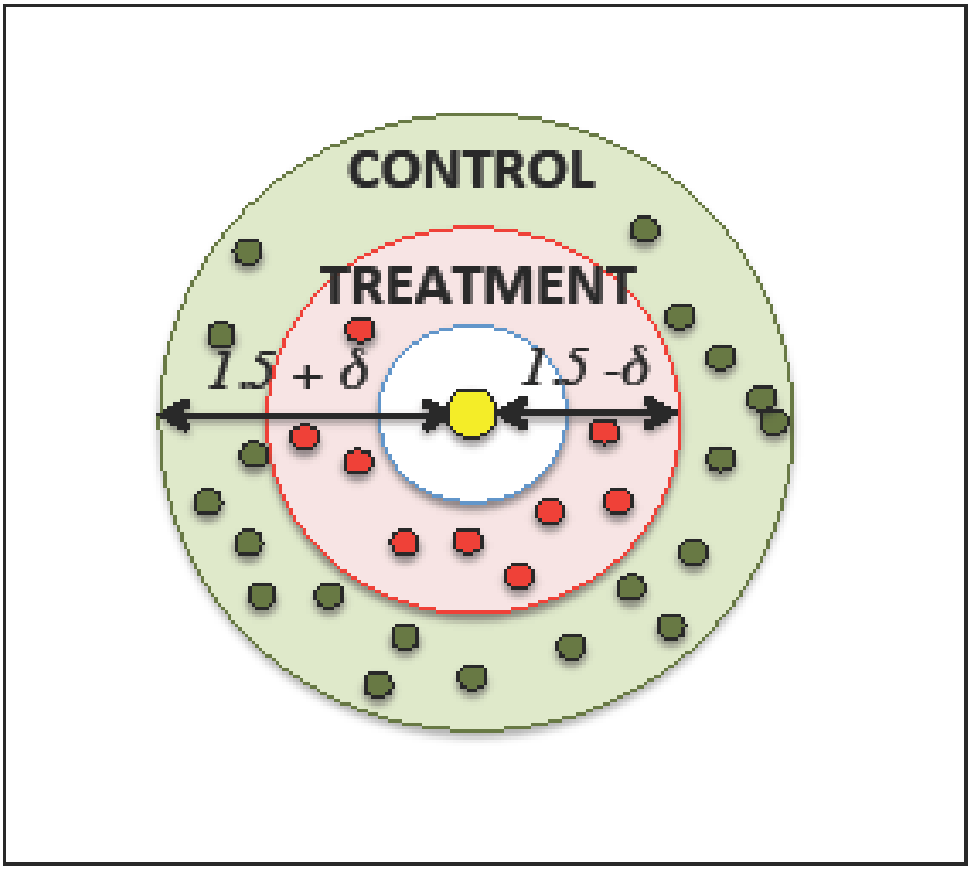


Figure 3

Treatment and Control Observations, δ = 1.5

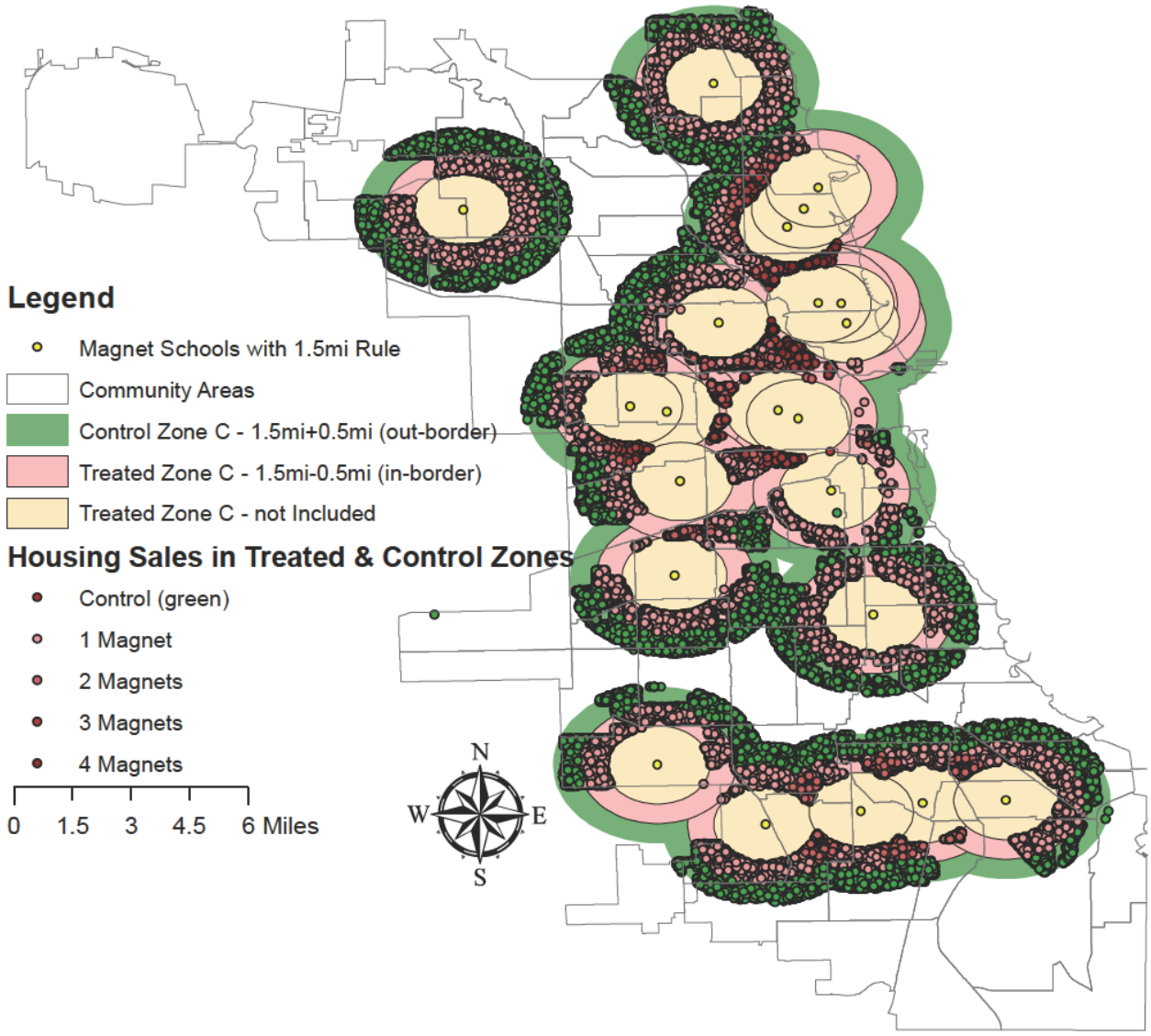


Figure 4

Treatment and Control Observations, δ = 0.5

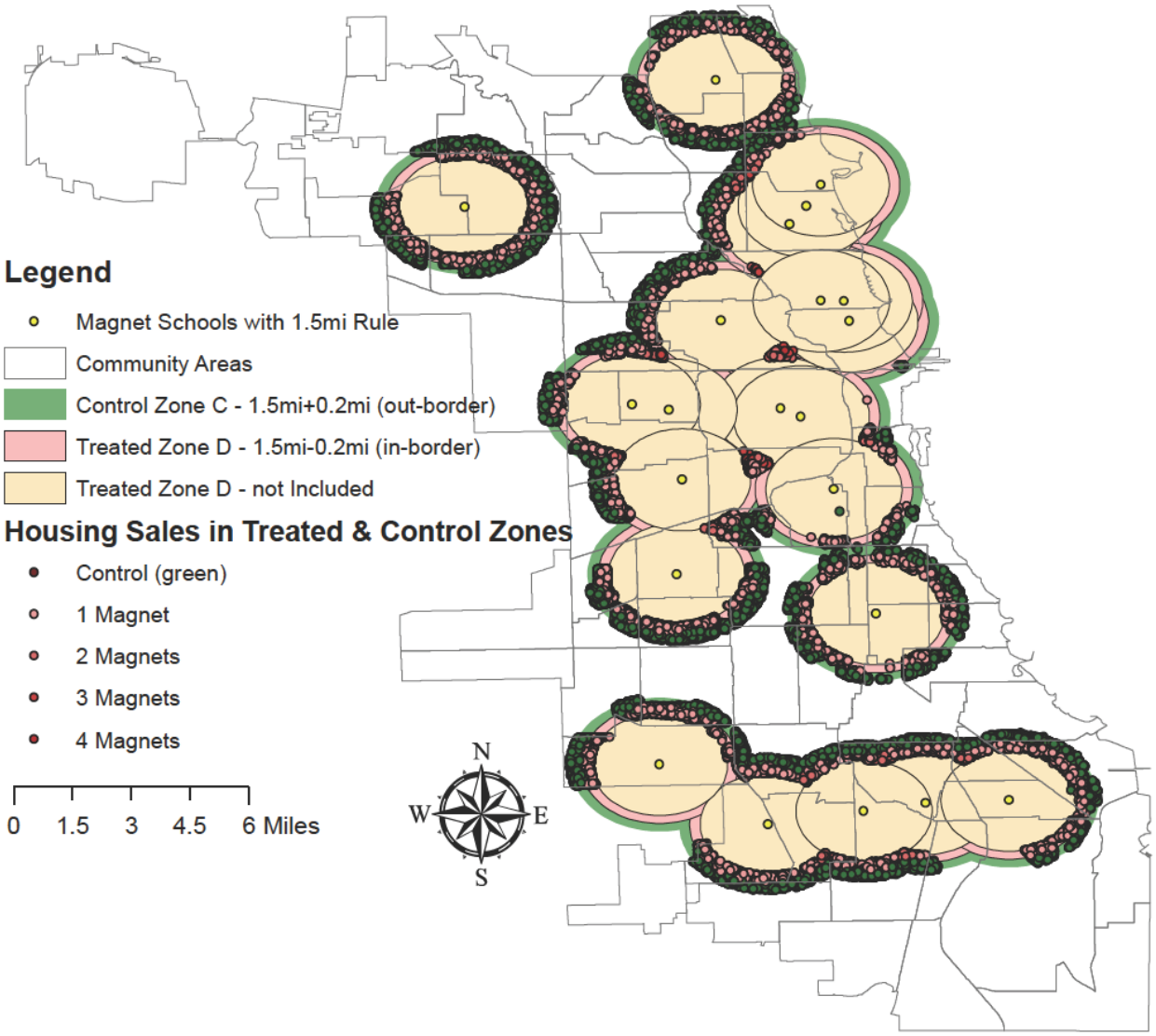


Figure 5

Treatment Intensity

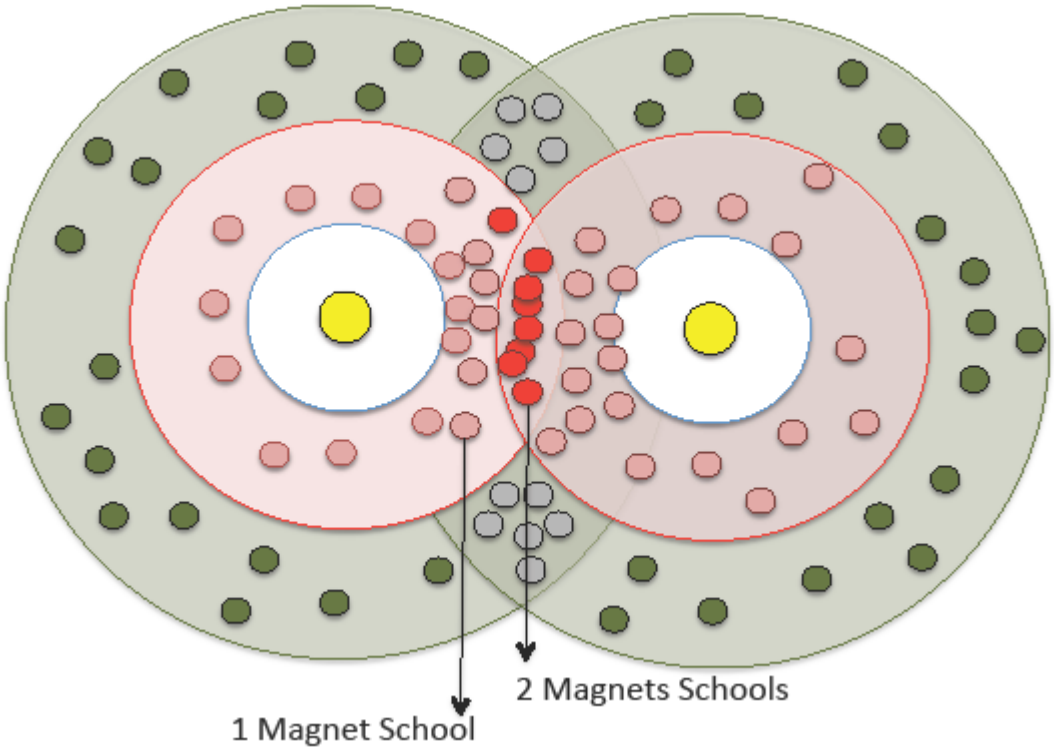
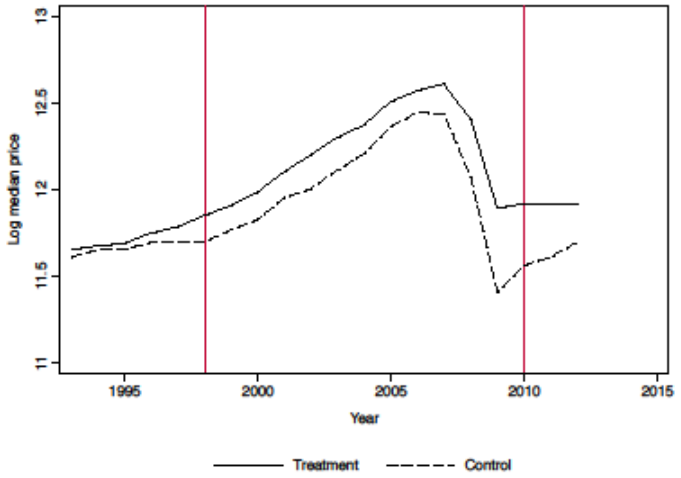


Figure 6

Median House Prices over Time



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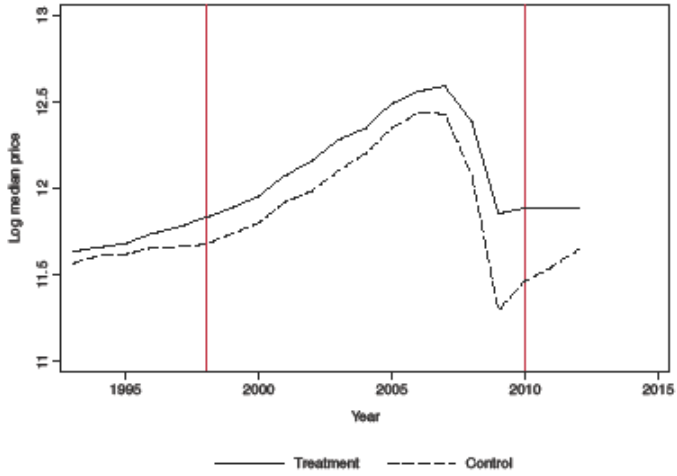

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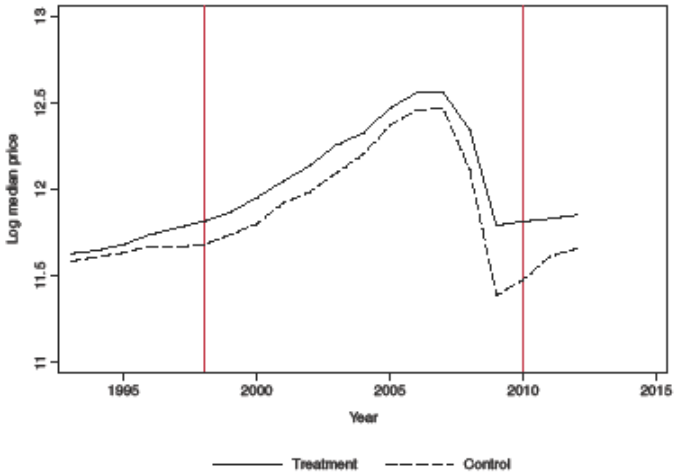
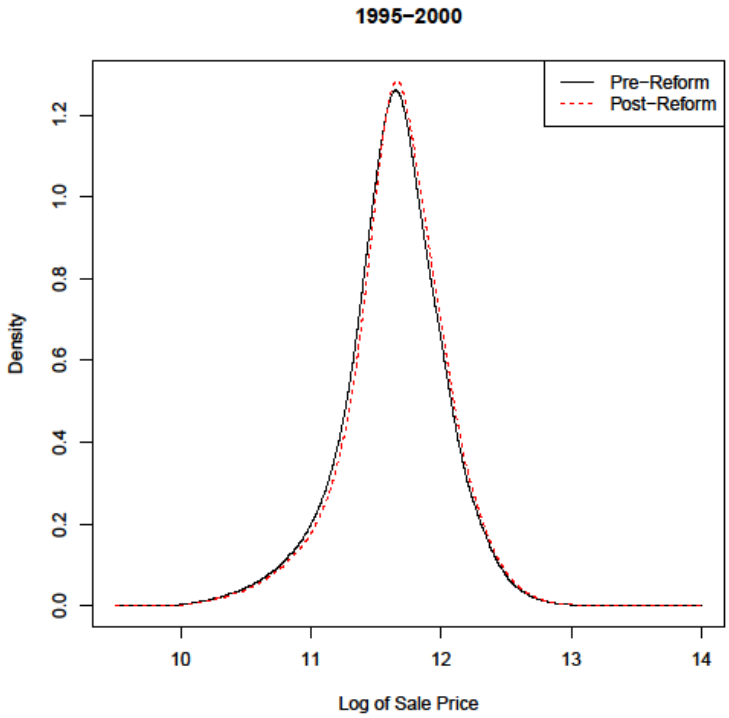


δ = 0.5

Figure 7

Quantile Estimates of Log Sale Price Distributions within 1.5 Miles of a Magnet School



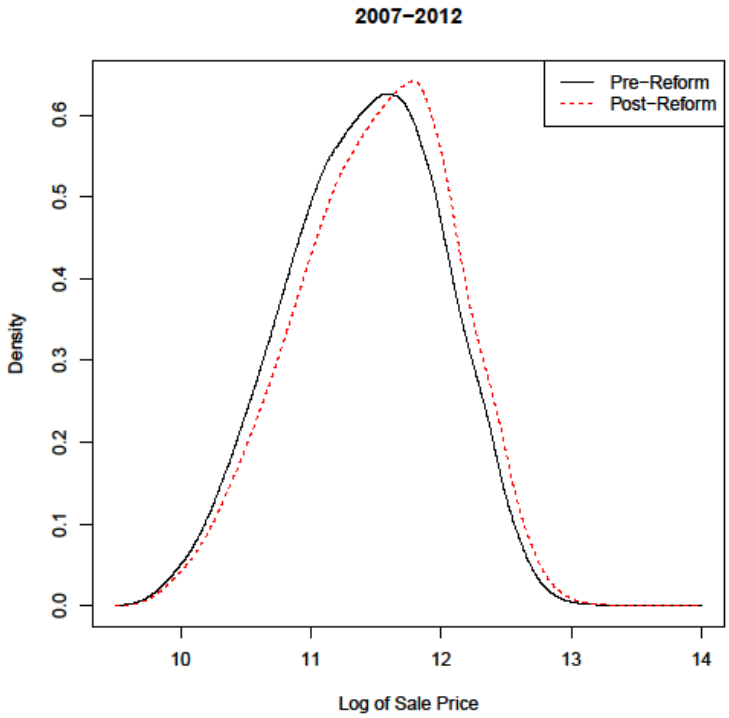
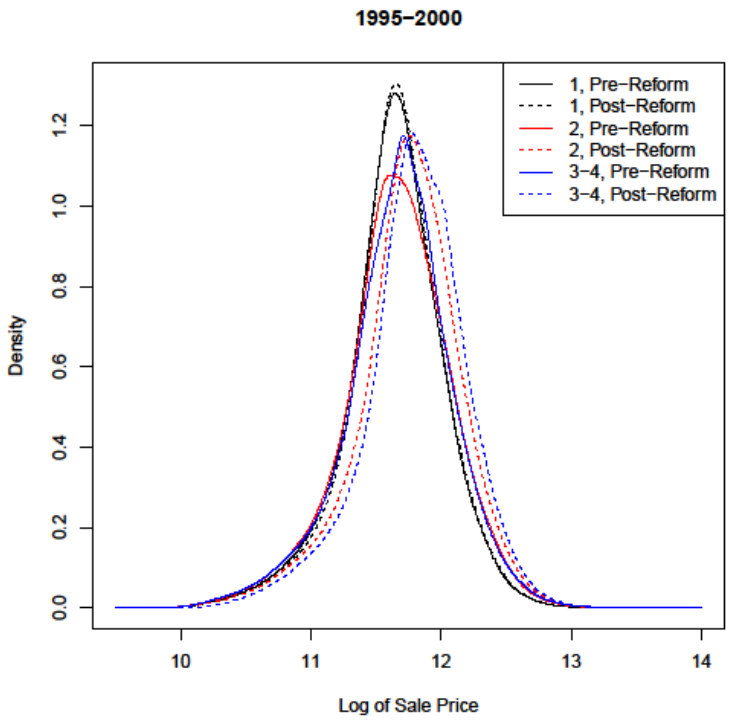
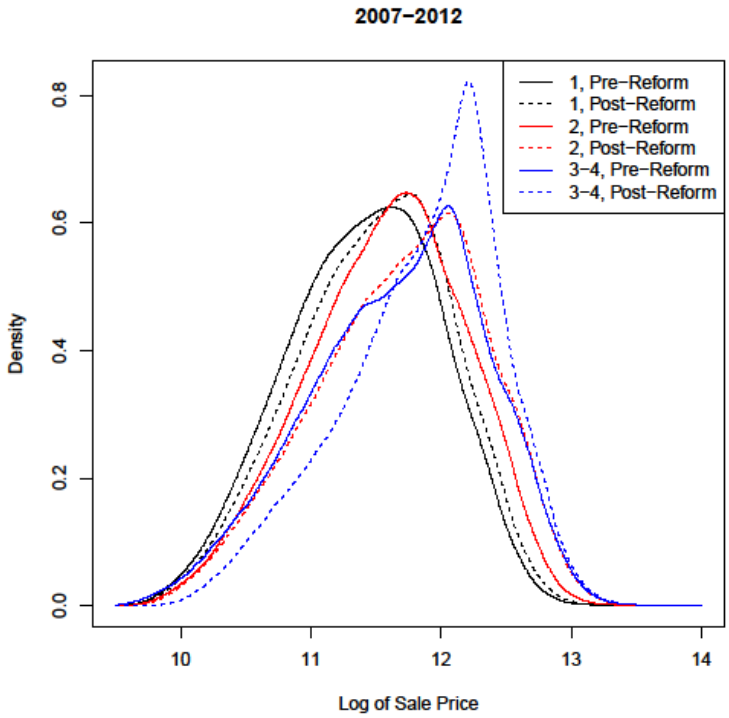


Figure 8

Quantile Estimates of Log Sale Price Distributions within 1.5 Miles of a Magnet School by Number of Nearby Schools